FORMULA SHEET:

UNIT : 1:

(1) CURVE FITTING

(1) $y = ax + b$

$$\sum y = a\sum x + nb$$
$$\sum xy = a\sum x^2 + b\sum x$$

(2) $y = ax^2 + bx + c$

$$\sum y = a\sum x^2 + b\sum x + nc$$
$$\sum xy = a\sum x^3 + b\sum x^2 + c\sum x$$
$$\sum x^2 y = a\sum x^4 + b\sum x^3 + c\sum x^2$$

(2) REGRESSION:

(1) $$\sigma^2 x = \frac{\sum (x-\bar{x})^2}{n} = \frac{\sum x_i^2}{n} - (\bar{x})^2$$
Variance

(2) Correlation coefficient:

$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{n \cdot \sigma x \cdot \sigma y} \qquad -1 \le r \le +1$$

Assume $z = x - y$

$\therefore$

$$r = \frac{\sigma^2 x + \sigma^2 y - \sigma^2 z}{2\sigma x \cdot \sigma y}$$

(3) Lines of Regression:

Point of intersection is mean

(i) $$y - \bar{y} = r\left(\frac{\sigma y}{\sigma x}\right)(x - \bar{x})$$ Regression of y on x

(ii) $$x - \bar{x} = r\left(\frac{\sigma x}{\sigma y}\right)(y - \bar{y})$$ Regression of x on y

(iii) $$\tan\theta = \left(\frac{\sigma x \cdot \sigma y}{\sigma^2 x + \sigma^2 y}\right)\left(\frac{1 - r^2}{r}\right)$$

(iv) $$r = \sqrt{\underset{(i)}{\text{coefficient of } x} \times \underset{(ii)}{\text{coefficient of } y}}$$

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④ MULTIPLE REGRESSION:

$$\sum y = na_0 + a_1 \sum x_1 + a_2 \sum x_2$$
$$\sum x_1 y = a_0 \sum x_1 + a_1 \sum x_1^2 + a_2 \sum x_1 x_2$$
$$\sum x_2 y = a_0 \sum x_2 + a_1 \sum x_1 x_2 + a_2 \sum x_2^2$$

⑬ PROBABILITY:

① DISCRETE PROBABILITY DISTRIBUTION:

(i) $P(x_i) \geq 0$

(ii) $\sum P(x_i) = 1$

(iii) Mean, $\mu = E(x) = \sum x_i \cdot P(x_i)$

(iv) Variance, $\sigma^2 = \sum (x-\mu)^2 \cdot P(x) = E(x^2) - \mu^2$

(v) $P(x < t) = \sum_{0}^{t} P(x)$

② CONTINUOUS PROBABILITY DISTRIBUTION:

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) \cdot dx = 1$

(iii) $\mu = E(x) = \int_{-\infty}^{\infty} x \cdot f(x) \cdot dx$

(iv) $E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot f(x) dx$

(v) $\sigma^2 = E(x^2) - \mu^2$

③ BINOMIAL DISTRIBUTION:

(i) $P(x) = {}^nC_x \cdot p^x \cdot q^{n-x}$, $p + q = 1$

(ii) mean $= np$

(iii) variance $= npq$

whole number → $\Gamma(n+1) = n!$; $\Gamma(\tfrac{1}{2}) = \sqrt{\pi}$

fraction → $\Gamma(n+1) = n.\Gamma(n)$

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④ POISSON DISTRIBUTION :

(i) n is very large , $n \to \infty$

p is very small , $p \to 0$

* $$p(x) = \frac{e^{-\lambda} \cdot \lambda^{x}}{x!}$$

(ii) $\mu = \sigma^2 = \lambda = np$

UNIT : 2 :

① PROBABILITY DISTRIBUTIONS :

① EXPONENTIAL DISTRIBUTION

(i) PDF , $$f(x) = \begin{cases} \alpha \cdot e^{-\alpha x} & , x > 0 \quad , \alpha > 0 \\ 0 & , \text{otherwise} \end{cases}$$

(ii) $\mu = 1/\alpha = \sigma$

(iii) $\sigma^2 = 1/\alpha^2$

② UNIFORM DISTRIBUTION :

(i) $$f(x) = \begin{cases} 1/b-a & , a \le x \le b \\ 0 & , \text{otherwise} \end{cases}$$

(ii) $\mu = \frac{a+b}{2}$ (iii) $\sigma^2 = \frac{(a-b)^2}{12}$

③ GAMMA DISTRIBUTION :

(i) $$P(x) = \begin{cases} \frac{x^{\alpha-1} \cdot e^{-x/\beta}}{\beta^{\alpha} \cdot \Gamma(\alpha)} & \\ 0 & \end{cases}$$

$= \cancel{\frac{\beta \cdot e^{-\beta x} \cdot (\beta x)^{\alpha-1}}{\Gamma(\alpha)}}$, $\alpha, \beta > 0$, $x \geq 0$

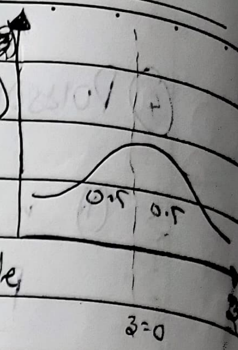
~~, otherwise~~

~~$\mu = \frac{\alpha}{\beta}$, $\sigma^2 = \frac{\alpha}{\beta^2}$~~

(ii) $\mu = \alpha\beta$, $\sigma^2 = \alpha\beta^2$

(4) Normal Distribution:

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$$



Standard distribution function: standard normal variable, $z = \frac{x-\mu}{\sigma}$

$$\cancel{P(x)} = P(z_1 < z < z_2) = \frac{1}{\sqrt{2\pi}} \cdot \int_{z_1}^{z_2} e^{-\frac{z^2}{2}} \, dz$$

$\mu = 0$, $\sigma^2 = 1 \cdot N$

(12) Joint Probability Distribution:

① Discrete — ② Continuous:

(i) $P(X=x_i, Y=y_j) = P_{ij} \geq 0$ — $f(x,y) \geq 0$

(ii) $\sum_i \sum_j P_{ij} = 1$ — $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dy\,dx = 1$

(iii) $P(x_i) = \sum_j P_{ij}$ — $f_1(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy$

(iv) $P(y_j) = \sum_i P_{ij}$ — $f_2(y) = \int_{x=-\infty}^{\infty} f(x,y)\,dx$

(v) $E(XY) = \sum_i \sum_j x_i y_j P_{ij}$ — $E(XY) = \int_{x=-\infty}^{\infty}\int_{y=-\infty}^{\infty} xy \cdot f(x,y)\,dy\,dx$

(vi) $\mu_x = \sum x_i P(x_i)$ — $\mu_x = \int_{-\infty}^{\infty} x \cdot f_1(x)\,dx$

$\mu_y = \sum y_i P(y_i)$ — $\mu_y = \int_{-\infty}^{\infty} y \cdot f_2(y)\,dy$

(vii) Covariance $(X, Y) = E(XY) - \mu_x \cdot \mu_y$

(viii) $\sigma_x^2 = E[x^2] - \mu_x^2$ — $\sigma_y^2 = E[y^2] - \mu_y^2$

(ix) Correlation, $\rho(XY) = \frac{\operatorname{Cov}(XY)}{\sigma_x \cdot \sigma_y}$

(x) $P(Y|X) = \frac{P(X=x, Y=y)}{P(x)}$, $P(X|Y) = \frac{P(X=x, Y=y)}{P(y)}$

Unit : 3 : Formula Sheet:

1) 1) (i) Probabilistic Vector : $\sum v_i = 1 \quad v_i \geq 0$

(ii) Regular Stochastic Matrix :

a) for some $m > 0 \Rightarrow P^m : P_{ij} > 0$

b) $VP = V$

c) $P, P^2, P^3 \ldots P^n$ all approach V

2) Markov Process :

(i) $tpm = P = \begin{bmatrix} P_{00} & \cdots & P_{0n} \\ \vdots & \ddots & \\ P_{m0} & \cdots & P_{mn} \end{bmatrix}$

(ii) $P_{ij} \rightarrow$ Probability of going from i^{th} to j^{th} state

(iii) $P_{ij}^{(n)} \rightarrow n^{th}$ step/stage transition probability

(iv) $P^{(n)} = P^{(0)} P^n$

2) 1) $(M|M|1 : \infty : FIFO)$

1) Traffic Intensity : $\rho = \frac{\lambda}{\mu}, \quad \lambda < \mu$

2) $L_s = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1-\rho} \qquad L_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{\rho^2}{1-\rho} = L_s \times \rho$

$W_s = \frac{1}{\mu - \lambda} \qquad W_q = \frac{\lambda}{\mu(\mu-\lambda)}$

$L_s = \lambda \cdot W_s \qquad L_q = W_q \cdot \lambda$

3) $P_0 = 1 - \frac{\lambda}{\mu} = 1 - \rho$

$P(n > 1) = \rho^2$ (not-empty queue)

$P(n > k) = \rho^{k+1}$ (number of customers exceed k)

$P_n = (1-\rho)\rho^n = P_0 \cdot \rho^n$ (n customers in the system)

4) $P(W_q > t) = \rho \cdot e^{-(\mu-\lambda)t}$

② $(M|M|1 : k|FIFO)$

1) λ may not necessarily be less than μ

2) $L_s = \begin{cases} \dfrac{\rho}{1-\rho} - \dfrac{(k+1)\rho^{k+1}}{1-\rho^{k+1}}, & \rho \neq 1 \\ \dfrac{k}{2}, & \rho = 1 \end{cases}$

3) $\lambda' = \mu(1-P_0)$

4) $L_q = L_s - \dfrac{\lambda'}{\mu}$

5) $\cancel{P_n = \begin{cases} \dfrac{1-\rho^n}{1-\rho^{k+1}}, & \rho \neq 1 \\ \dfrac{1}{k+1}, & \rho = 1 \end{cases}}$

8) $P(N \geq n) = \dfrac{(1-\rho)\rho^n}{1-\rho^{k+1}}$

6) $W_s = \dfrac{L_s}{\lambda'}$

7) $W_q = \dfrac{L_q}{\lambda'}$

5) $P_0 = \begin{cases} \dfrac{1-\rho}{1-\rho^{k+1}}, & \rho \neq 1 \\ \dfrac{1}{k+1}, & \rho = 1 \end{cases}$

9) $P_n = \rho^n \cdot P_0$
$= \rho^n \left(\dfrac{1-\rho}{1-\rho^{k+1}} \right)$

(3) $(M/M/S : \infty / FIFO)$

1) $P_0 = \left[\left[\sum_{n=0}^{s-1} \frac{1}{n!} \rho^n\right] + \frac{\rho^s}{s!\,(1-\lambda/\mu s)}\right]^{-1}$

2) $P_n = \begin{cases} \dfrac{\rho^n \cdot P_0}{n!}, & n < s \\ \dfrac{\rho^n \cdot P_0}{s! \cdot s^{n-s}}, & n > s \end{cases}$

3) $L_q = \dfrac{P_0}{s \cdot s!} \cdot \dfrac{\rho^{s+1}}{(1-\rho/s)^2}$

4) $L_s = L_q + \rho$

5) $W_q = \dfrac{L_q}{\lambda}$ 6) $W_s = W_q + \dfrac{1}{\mu}$

7) $P(N \geq s) = \dfrac{\rho^s \cdot P_0}{s!} \left(\dfrac{1}{1-\rho/s}\right)$

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Unit : 4: Formula Sheet:

(1) Z-Test : For Large Sample $(n \geq 30)$

(i) standard error $= \frac{\sigma}{\sqrt{n}}$

distribution is non-normal + large sample size

(ii) $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ (iv) $\mu = np$ (v) $\sigma = \sqrt{npq}$

(iii) $-z_1 \leq Z \leq z_1 \Rightarrow$ Confidence Limit, $CL = \bar{x} \pm z_1 \cdot \frac{\sigma}{\sqrt{n}}$

	at 5%	at 1%
Two-tailed	1.96	2.58
One-tailed	1.645	2.33

(vi) When $\bar{x}$ is not available:

$$Z = \frac{x - \mu}{\sigma}$$

(vii) for different population:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

(viii) for same population, multiple algorithms:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

(ix) When σ is unknown:

$$S = \sqrt{\frac{1}{n}\sum (x-\bar{x})^2}$$

(x) When σ_1 & σ_2 are unknown for two populations:

$$S = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2}}$$

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{S\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

12) t-TEST : SMALL SAMPLE ($n < 30$)

(i) $t = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{\bar{x}-\mu}{s\sqrt{n-1}}$ for $(n-1)$ degree of freedom

(ii) $CL = \bar{x} \pm t_\alpha \cdot \frac{\sigma}{\sqrt{n}} = \bar{x} \pm t_\alpha \frac{s}{\sqrt{n-1}}$

(iii) When testing for deviation:

$$t = \frac{\bar{d}}{s/\sqrt{n-1}}, \quad s = \sqrt{\frac{1}{n}\sum(d-\bar{d})^2}, \quad \mu = 0$$

(iv) Testing for significant difference of sample means:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$ for (n_1+n_2-2) d.f

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(v) When σ isn't known:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad s = \sqrt{\frac{n_1 \cdot s_1^2 + n_2 \cdot s_2^2}{n_1 + n_2 - 2}}$$

for $(n_1 + n_2 - 2)$ d.f

13) F-TEST : Test of significance in difference of VARIANCES :

$$F = \frac{\sigma_1^2}{\sigma_2^2} = \frac{s_1^2}{s_2^2} \quad \text{for } (n_1 - 1, n_2 - 1) \text{ d.f}$$

14) CHI-SQUARE (χ^2) TEST

① Goodness of fit:

$$\chi^2 = \sum_{i=1}^{n} \frac{(O_i - E_i)^2}{E_i}$$

for
- $(n-1)$ df for binomial distribution
- $(n-2)$ df for Poisson distribution
- $(n-3)$ df for Normal distribution

② Test of Independence : Contigency Table

a	b	$R_1 = (a+b)$
c	d	$R_2 = (c+d)$
$C_1 = (a+c)$	$C_2 = (b+d)$	N

$$E_i = \frac{R_i \times C_i}{N}$$

$O_i = a, b, c, d \ldots$

$$df = (n_R - 1)(n_C - 1)$$

(number of row − 1) (number of cols − 1)